Article

Momentum Contrastive Learning on 3D Local Features

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Abstract: Self-supervised contrastive learning has demonstrated remarkable effectiveness in learning visual representations without labeled data, yet its application to 3D local feature learning from point clouds remains underexplored. Existing methods predominantly focus on complete object shapes, neglecting the critical challenge of recognizing partial observations commonly encountered in real-world 3D perception. We propose a momentum contrastive learning framework specifically designed to learn discriminative local features from randomly sampled point cloud regions. By adapting the MoCo architecture with PointNet++ as the feature backbone, our method treats local clusters as fundamental contrastive learning units, combined with carefully designed augmentation strategies including random dropout and translation. Experiments on ShapeNet demonstrate that our approach effectively learns transferable local features, requiring clusters of at least 30% object size, and achieves comparable downstream classification accuracy while reducing training time by 16%.

Keywords: 3D point cloud; self-supervised learning; contrastive learning; momentum encoder; local feature representation

1. Introduction

In recent years, 3D computer vision has advanced significantly alongside the development of 2D vision tasks, enabling essential applications including 3D object detection, classification, semantic segmentation, and reconstruction. However, unlike structured and continuous 2D images, 3D data presents unique challenges stemming from its inherent properties—unordered point sets, sparse spatial distributions, and irregular structures. These characteristics lead to substantial computational costs, labor-intensive annotation requirements, and inevitable information loss during data acquisition.

Due to the exclusive properties of 3D data, different types of data structure are used for solving 3D vision tasks [1–9]. Point clouds represent the most fundamental format for capturing real-world 3D geometry. As shown in Figure 1, a point cloud is simply a collection of 3D coordinates in Euclidean space, where each point may contain additional attributes such as color, intensity, reflectivity, or normal information [10]. This unstructured nature, while preserving geometric fidelity, introduces significant obstacles for traditional supervised learning approaches [11–14]. The high dimensionality of point cloud data, combined with the scarcity of large-scale labeled datasets, makes it impractical to rely solely on supervised training mechanisms for extracting meaningful representations.

Self-supervised learning (SSL) has emerged as a promising solution, achieving remarkable success in natural language processing [15–17] and 2D vision tasks [18–20]. By learning from unlabeled data through carefully designed pretext tasks, SSL methods can

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extract useful representations without human annotations. Contrastive learning, a particularly effective SSL paradigm, encourages different augmented views of the same input to produce similar representations while separating representations from different inputs. For 3D point clouds, this approach leverages various data augmentation techniques to create multiple views, enabling the model to learn robust features through self-supervision [21]. This paradigm not only reduces computational costs [22] but also circumvents the expensive annotation process by generating pseudo-labels through clustering methods such as memory bank [18], while preventing information loss through well-designed contrastive objectives [23].

Despite these advances, existing 3D self-supervised contrastive learning methods predominantly focus on learning global shape representations of complete objects [24,25]. This global-only approach overlooks a critical aspect of real-world 3D perception: objects are frequently observed as partial views due to occlusion, sensor limitations, or viewpoint constraints. Recent study on 3D classification and segmentation have shown that model performance significantly degrades by 30-60% in accuracy when tested on partially visible or occluded objects, even when trained on complete shapes [26]. To address the importance of local information, some recent works have explored part-based learning strategies. DHGCN [27] voxelizes point clouds into parts to discover their contextual relationships, while LAA [28] introduces a local-global attention module that strengthens local cues. Although both methods emphasize locality, they depend on predefined part grouping or architectural changes, and they do not treat local regions themselves as the primary training unit in contrastive learning. Therefore, current SSL methods lack the ability to learn discriminative features directly from geometric local regions, which are essential for recognizing objects under realistic conditions.

To address this gap, we propose a modified momentum contrastive learning framework specifically designed for learning 3D local feature representations. Building upon the Momentum Contrast (MoCo) architecture [29], a pioneering work that introduced momentum-based contrastive mechanisms for 2D tasks, we adapt this simple yet effective structure for 3D point cloud processing. Unlike previous methods that rely on predefined part groupings, our approach treats randomly sampled local regions as the fundamental training units in contrastive learning. We employ PointNet++ as the feature learning backbone and introduce a random local clustering method combined with carefully designed 3D data augmentation strategies. By randomly sampling and learning from local parts of 3D objects, our framework explicitly models the partial observation scenarios commonly encountered in real-world 3D scenes, thereby learning more robust and transferable local representations that directly capture geometric information from local regions.

We conduct comprehensive experiments on ShapeNet [30,31] to evaluate our proposed method. Our investigation focuses on three key aspects: (1) the relationship between local cluster size and feature learning capability, (2) the effectiveness of different data augmentation methods for local feature training, and (3) the transferability of learned features to downstream 3D object classification tasks. This work makes the following contributions:

- we present a momentum contrastive learning framework tailored for 3D local feature representation learning, addressing the problem of self-supervised learning directly from partial point cloud observations without predefined part structures.
- We systematically analyze the impact of local cluster size and data augmentation strategies on local feature learning, establishing that local clusters comprising at least 30% of object points can be effectively learned through momentum contrastive training.

- We demonstrate through downstream classification experiments that models pre-trained with our local feature learning approach achieve comparable final accuracy to models trained from scratch, while reducing training time by approximately 16%, thereby validating the effectiveness of our learned local representations for transfer learning.

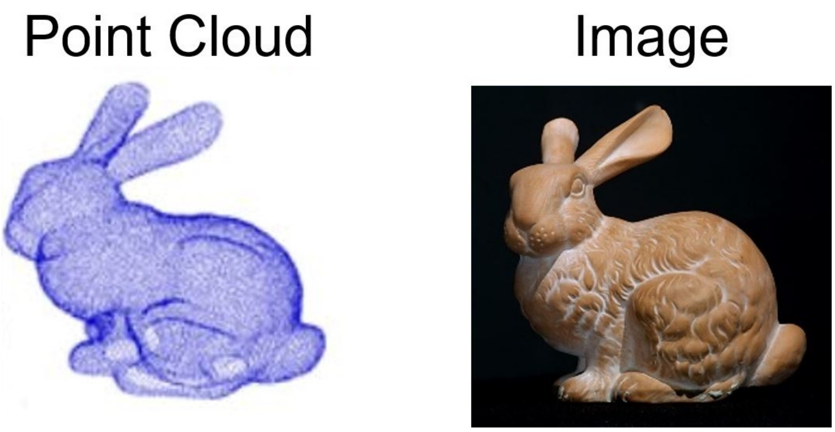

Figure 1. Samples of Point Cloud and Image.

2. Related Work

2.1. *Self-supervised Training*

Self-supervised training has achieved remarkable success in natural language processing tasks such as GPT [15,17] and BERT [16], and has increasingly been adopted for vision tasks. Early self-supervised approaches for images imposed simple variations on inputs and extracted features by recovering the original data. These include semantic-label-based methods [32,33] and cross-modal-based methods [34–36]. However, these methods incur extra computational costs and memory overhead for generating labels and learning from other modalities.

Recently, contrastive learning has gained significant attention due to its ability to rely solely on the dataset itself by generating positive and negative samples through data augmentation. Wu et al. store previous feature encodings in a memory bank [18], while Momentum Contrast (MoCo) [29] extends this approach with an updating queue for negative sample features and a momentum update encoder, surpassing ImageNet-supervised counterparts in multiple detection and segmentation tasks. Other advances [19,20] have further demonstrated that self-supervised training is highly effective for 2D vision tasks.

For 3D vision tasks, Xie et al. proposed PointContrast [24], which augments 3D scenes with rotations and color transformations before contrasting transformed point clouds using a contrastive loss function. To address the limitation of requiring multiple views, Zhang et al. presented DepthContrast [25], which learns representations from depth maps using only single-view point cloud data. It constructs two augmented versions through data augmentation, with format-specific encoders generating spatial features from either point or voxel representations, ultimately obtaining global features for instance discrimination.

However, these works focus on pretraining with complete object shapes, thereby neglecting local part information. Recent studies have begun to highlight the importance of local information in point cloud understanding. DHGCN [27] voxelizes point clouds into parts and predicts hop distances between parts to learn their contextual relationships, improving structural reasoning across point parts. LAA [28] introduces a local-global attention module that strengthens local cues while maintaining global context in transformer-based pretraining, thereby improving downstream performance. Work in other visual domains [37] has similarly shown benefits from structured region perception. Although both methods emphasize locality, they depend on predefined part grouping or architectural modifications and do not treat local regions themselves as the primary training unit in contrastive learning. Moreover, they do not systematically examine how the size and variation of local regions affect learned features. Despite these advances, existing 3D self-supervised methods predominantly learn from complete shapes and lack explicit mechanisms for learning discriminative features directly from randomly sampled local geometric regions, which are critical for handling partial observations in real-world scenarios [27,28,38]. While diffusion-based approaches have shown promise in modeling motion trajectories for sequential tasks [39], their application to self-supervised point cloud feature learning remains unexplored.

Our work takes a different approach: we sample local parts directly as contrastive learning inputs and systematically investigate how local region characteristics affect feature representation, providing a simple and direct method to learn local geometric structure without relying on handcrafted part definitions or complex architectural modifications.

2.2. Point Cloud Encoder

Point cloud encoders have emerged as essential components in 3D vision tasks due to the unique challenges of processing unstructured 3D data. Point-based models capture representations at the point-wise level. PointNet [40] pioneered the direct processing of raw point clouds by learning spatial encodings of individual points and aggregating them into a global signature. However, it suffers from the loss of local geometric information. To address this limitation, PointNet++ [41] introduced a hierarchical structure that extracts features while considering both local and global information through multiple stages of sampling and grouping. This hierarchical design with explicit local neighborhood aggregation makes it particularly well-suited for processing local point patches, as it can effectively capture fine-grained geometric patterns within localized regions.

Building upon PointNet++, subsequent works have proposed various enhancements. Qian et al. introduced PointNeXt [42], which incorporates a separable MLP and an inverted residual bottleneck design. Hu et al. proposed RandLA-Net [43], which employs random point sampling and increases the receptive field through an efficient local feature aggregation module. Ma et al. presented PointMLP [44], a residual model that applies a geometric affine module instead of a local geometry extractor.

Since the introduction of the Transformer [45], many encoders have adopted this architecture for point cloud processing. Guo et al. proposed Point Cloud Transformer (PCT) [46], which provides permutation invariance and brings the transformer to point cloud feature learning. Pang et al. developed Point-MAE [47], a masked autoencoder that addresses local information leakage.

While these works propose point cloud encoders primarily for supervised training, our work leverages the point cloud encoder within a self-supervised training framework specifically designed for extracting local features. To efficiently represent 3D local clusters, we adopt PointNet++ as our feature learning backbone due to its hierarchical structure and effective local neighborhood capturing capabilities, which are particularly advantageous for processing local point patches. Furthermore, the learned features can be directly transferred to downstream tasks such as 3D object classification and detection.

3. Method

To learn discriminative 3D local features through contrastive learning, we propose a modified momentum contrastive pretraining architecture specifically designed for local point cloud representations. This section presents the overall framework architecture, including the momentum contrastive learning mechanism with its contrastive loss and momentum update strategy, the point cloud feature learning backbone, and the data processing pipeline comprising augmentation methods and local cluster sampling.

3.1. Momentum Contrastive Learning Structure

The proposed framework is illustrated in Figure 2. The architecture consists of two primary modules: a data augmentation module and a local feature learning module. The data augmentation module processes input 3D point clouds into transformed local clusters that serve as queries and keys. The local feature learning module then learns discriminative local representations from these query and key clusters through contrastive learning.

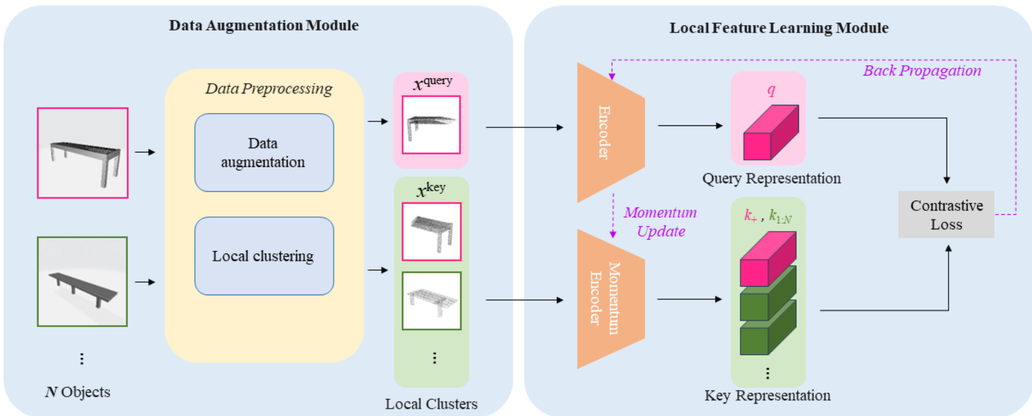

Figure 2. Momentum Contrastive Learning Structure for 3D Local Features.

The complete data flow proceeds as follows: First, an input point cloud undergoes data augmentation transformations. Second, the KD-Tree algorithm samples local clusters from the augmented point cloud. Third, these local clusters are processed by the query encoder and momentum encoder to generate feature representations. Finally, the InfoNCE loss is computed to train the encoders. This step-by-step pipeline ensures that local geometric information is effectively captured and learned through contrastive objectives.

Data Augmentation Module. The data augmentation module takes a complete point cloud as input and outputs a batch of transformed local clusters. Within each batch, the local clusters are designated as queries and keys. A query-key pair is considered positive if both clusters originate from the same CAD model; otherwise, they form negative pairs. The data augmentation methods generate local clusters with diverse geometric variations, enabling the momentum contrast framework to learn robust local features. The local clustering method randomly samples regions from the augmented point clouds, producing local clusters that capture different geometric patterns.

Local Feature Learning Module. This module learns local representations from query and key clusters while updating the encoders through momentum contrast. To ensure feature consistency during contrastive learning, both the query encoder and momentum encoder share the same architectural structure. During training, the query encoder is updated via standard back-propagation, while the momentum encoder is updated through a momentum mechanism, resulting in models with identical architecture but different parameters. The contrastive loss function encourages the query features to match their corresponding positive key features while remaining distinct from negative key features, thereby training the encoders to learn discriminative local representations.

3.1.1. Contrastive Loss

Local clusters are encoded into query and key features by the query encoder and momentum encoder, respectively. Given a batch size of N samples, the encoder produces one query feature q and a set of key features $\{k_+, k_1, k_2, \ldots, k_{N-1}\}$. These key features constitute a dictionary for contrastive learning. The positive pair (q, k_+) for $i = 1, 2, \ldots, N-1$ come from different point clouds. To encourage the query feature q to be similar to its positive key k_+ and dissimilar to negative keys, we employ the InfoNCE loss [48]:

$$\mathcal{L}_q = -\log \frac{exp(q \cdot k_+/\tau)}{exp(q \cdot k_+/\tau) + \sum_{i=1}^{N-1} exp(q \cdot k_i/\tau)}, \tag{1}$$

where τ is a temperature hyperparameter that controls the smoothness of the softmax distribution. We set $\tau = 0.07$ following standard practice in contrastive learning. This loss function can be interpreted as the negative log-likelihood of a $(K+1)$-way softmax classifier that attempts to identify the positive key k_+ from the query q. By minimizing this loss, the model learns to pull together local features from the same point cloud in the latent space while pushing apart features from different point clouds. This unsupervised objective enables the encoders to capture informative local geometric patterns through contrastive learning on local point cloud regions.

3.1.2. Momentum Update

The contrastive loss trains the encoders to produce discriminative local features for both query and key clusters. The dictionary of key features is maintained as a queue that is progressively updated: as each new mini-batch is processed, its key features are enqueued while the oldest mini-batch is dequeued. This queue mechanism maintains a large and relatively current set of negative samples for contrastive learning. We set the queue size to 65,536 to provide sufficient negative samples for effective contrastive learning.

However, applying standard back-propagation to update both encoders would cause inconsistencies in the key representations stored in the queue. Specifically, if the key encoder parameters were updated as rapidly as the query encoder, the key features in the queue would be generated by encoders with vastly different parameters, undermining the coherence of the contrastive learning process. To address this issue, we employ a momentum update mechanism for the momentum encoder:

$$\theta_k \longleftarrow m\theta_k + (1-m)\theta_q, \tag{2}$$

where θ_q denotes the parameters of the query encoder f_q, θ_k denotes the parameters of the momentum encoder f_k, and $m \in [0, 1)$ is the momentum coefficient. This momentum update ensures that f_k evolves slowly based on f_q, rather than through direct back propagation. By maintaining a slowly evolving momentum encoder, the key features in the queue remain relatively consistent despite being generated at different training iterations. Following MoCo [29], we set $m = 0.999$, which empirically provides the best balance between stability and adaptability.

3.2. Point Cloud Feature Learning Backbone

To effectively learn informative local features from point clouds, we adopt PointNet++ [41] as our feature learning backbone. PointNet++ is particularly well-suited for local feature learning due to its hierarchical structure that explicitly captures multi-scale local geometric patterns through successive sampling and grouping operations. This design enables the network to aggregate information from local neighborhoods at multiple scales, making it ideal for extracting discriminative features from local point cloud clusters.

The network takes XYZ coordinates of 3D points as input and employs a U-Net-like architecture [49] with four encoding layers for feature extraction and downsampling, followed by two decoding layers for feature aggregation and upsampling. In our implementation, each local cluster contains 1024 points, represented as a 1024×3 matrix of XYZ coordinates. The hierarchical sampling and grouping operations progressively reduce the point set size while increasing the feature dimensionality, allowing the network to capture both fine-grained local details and broader contextual information. The final layer produces C-dimensional per-point features for 128 sampled points after the aggregation process, where $C = 256$ in our experiments. These per-point features are then max-pooled to obtain a global feature vector for the local cluster, which serves as the input to the projection head for contrastive learning.

The explicit local-to-global hierarchical structure of PointNet++ naturally aligns with our objective of learning local geometric features. By processing local clusters through this backbone, the network learns to identify distinctive local geometric patterns while maintaining awareness of the broader spatial context within each cluster.

3.3. Data Processing

3.3.1. Data Random Augmentation Methods

To improve the robustness of learned local features and enable the model to handle various real-world scenarios, we apply three standard 3D augmentation methods: random input dropout, random rotation, and random translation. These augmentations generate diverse views of local clusters, encouraging the model to learn invariant local geometric features.

Random input dropout randomly removes 20-30% of points from each local cluster, simulating the partial occlusions and missing data frequently encountered in real-world 3D sensing. This augmentation forces the model to recognize local geometric patterns even when parts of the structure are absent.

Random rotation applies random rotations around the vertical axis (z-axis) within the range of 0-360°, generating local clusters with different orientations. This augmentation ensures that learned features are invariant to object orientation, which is crucial for recognizing local patterns regardless of viewpoint.

Random translation randomly shifts the local cluster positions within a range of ±0.2 meters along each axis, simulating viewpoint variations during 3D data acquisition. This augmentation encourages the model to focus on relative geometric relationships within local regions rather than absolute spatial positions.

These augmentation methods work together to create a rich variety of local cluster appearances from each input point cloud, enabling the momentum contrastive framework to learn robust and generalizable local geometric features. We conduct ablation studies to evaluate the individual contribution of each augmentation method to the overall performance.

3.3.2. Local Clustering Method

To sample local clusters from complete point clouds, we employ the K-dimensional Tree (KD-Tree) algorithm, which is a space-partitioning data structure that efficiently organizes k-dimensional points in a tree structure for fast range searches and nearest neighbor queries. The KD-Tree algorithm is particularly suitable for local cluster sampling due to its computational efficiency, which significantly accelerates the pretraining process.

The local clustering process proceeds as follows: First, we randomly select a center point from the augmented point cloud. Second, we use the KD-Tree to identify the K nearest neighbors of the center point, where K is set to 1024 to form a complete local cluster. This approach ensures that each local cluster captures a coherent local geometric region while the random center selection provides diverse local samples across the entire point cloud. By sampling local clusters in this manner, we can generate multiple local views from a single point cloud, each representing different geometric patterns and structural characteristics. This random sampling strategy enables the model to learn from varied local geometric configurations, enhancing its ability to recognize partial shapes and local patterns in downstream tasks.

4. Experiments

We conduct two stages of experiments to evaluate our proposed momentum contrastive pretraining framework for 3D local features. The first stage investigates pretraining effectiveness by examining how local cluster size and data augmentation methods influence the learning of local representations. The second stage evaluates the transferability of learned features through downstream 3D object classification tasks, comparing models with and without pretraining. Through these experiments, we demonstrate that our method effectively learns discriminative local features that accelerate downstream task training.

4.1. Dataset

We use ShapeNet [31] as our pretraining dataset. As illustrated in Figure 3, it is a collection of single-object CAD models containing 57,448 objects across 55 categories, developed by researchers from Stanford University, Princeton University, and the Toyota Technological Institute at Chicago. This dataset is widely adopted for 3D vision tasks and provides rich geometric information for computer graphics and vision research.

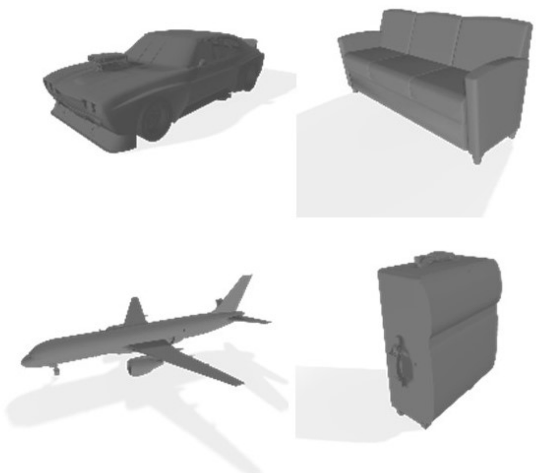

Figure 3. Samples of ShapeNet.

For pretraining experiments, we randomly select 30% of the ShapeNet dataset as our training set. This subset size is chosen to accelerate the pretraining process while maintaining sufficient diversity to validate our method's effectiveness across different local geometric patterns. Our primary objective is to demonstrate the viability of local feature learning through momentum contrastive pretraining rather than achieving maximum accuracy on the full dataset. For the downstream classification task, we select five representative object categories including table, chair, lamp, bench, and bookshelf. It focus the evaluation on how learned local features influence classification performance without introducing complexity from large-scale multi-class scenarios. This focused downstream setting allows us to isolate the effect of local feature pretraining on transfer learning efficiency.

Each local cluster is sampled to contain 1024 points from the input point cloud. We set the batch size to 64 and the learning rate to 0.007 for pretraining. All pretraining experiments are conducted for 200 epochs to ensure convergence.

4.2. Momentum Contrast Pretraining

We design two comparison experiments to systematically evaluate our momentum contrast pretraining method. The first experiment examines the relationship between local cluster size and feature learning capability, aiming to identify the minimum cluster size required for effective contrastive learning. The second experiment investigates the contribution of different data augmentation methods to local feature extraction.

4.2.1. Training with Different Partial Scales

This experiment evaluates how local cluster size affects the learning of local features through momentum contrast. We train models with local clusters ranging from 20% to 90% of the complete object, maintaining identical data augmentation strategies across all settings. The results enable us to determine whether momentum contrast is effective for learning 3D local features and to identify the minimum cluster size necessary for meaningful feature learning.

The results are presented in Table 1. We report two metrics: Acc@1 (top-1 classification accuracy, where the highest probability classification corresponds to the positive key) and Acc@5 (top-5 classification accuracy, where the positive key appears among the top five predictions). As shown in Table 1, both Acc@1 and Acc@5 increase as local cluster size grows, confirming that our contrastive learning framework can effectively learn from local features. Notably, clusters containing 33% to 90% of the object achieve similar performance levels (Acc@1 ranging from 44.71% to 51.68%), indicating that moderate-sized local regions contain sufficient geometric information for contrastive learning.

However, performance degrades dramatically when cluster size falls below 30%, with Acc@1 dropping to 25.62% at 30% and nearly 0% at 20% and 25% cluster size. This sharp decline suggests that clusters smaller than 30% lack sufficient local geometric information to form stable positive pair relationships in the contrastive learning framework. When local regions become too small, they may represent ambiguous or incomplete geometric patterns that cannot be reliably distinguished from negative samples, causing the contrastive learning objective to fail. Based on these findings, we conclude that local clusters must contain at least 30% of the complete object to enable effective momentum contrastive learning of local features.

Table 1. Comparison Result with Different Size of Local Cluster

Percentage of local cluster size	Loss	Acc@1 [%]	Acc@5 [%]
90%	8.05	44.71	50.57
80%	7.90	50.83	51.06
66%	8.03	51.68	51.68
50%	8.13	50.65	51.03
33%	8.00	50.59	50.62
30%	8.03	25.62	25.98
25%	8.52	0.31	1.24
20%	8.21	0.28	1.08

Table 2. Comparison Result with Different Data Augmentation Methods

Data augmentation methods	Loss	Acc@1 [%]	Acc@5 [%]
All	8.13	50.65	51.03
w/o random input dropout	8.34	0.59	1.45
w/o random rotation	8.19	49.85	50.98
w/o random translation	8.36	11.29	12.37

4.2.2. Training with Different Data Augmentation Methods

This experiment investigates the contribution of each data augmentation method to local feature learning by systematically removing individual augmentation techniques. We train models with local clusters containing 50% of the complete object and compare performance across different augmentation combinations. The results are shown in Table 2.

The results reveal that random input dropout and random translation are essential for effective local feature learning, while random rotation provides minimal benefit. When random input dropout is removed, performance collapses to near-zero (Acc@1 = 0.59%), indicating that this augmentation is critical for learning robust local features. This dramatic degradation suggests that dropout forces the model to recognize local geometric patterns even when parts of the structure are missing, simulating the partial occlusions encountered in real-world scenarios. Without this augmentation, the model may overfit to complete local clusters and fail to generalize to partial views.

Similarly, removing random translation significantly reduces performance. This finding suggests that translation augmentation helps the model focus on relative geometric relationships within local regions rather than absolute spatial positions, encouraging the learning of position-invariant local features. In contrast, removing random rotation has negligible impact on performance, likely because local patches exhibit weak geometric directionality—the intrinsic shape of a small local region is largely invariant to rotation around the vertical axis. These ablation results confirm that our augmentation strategy is well-designed, with dropout and translation playing complementary roles in promoting robust local feature learning.

4.3. Downstream Tasks

To further validate that our momentum contrastive pretraining method effectively learns transferable local features, we conduct downstream 3D object classification experiments comparing models with and without pretraining. The pretrained model is obtained using the optimal configuration identified in the pretraining experiments: local clusters containing 50% of the object and all three augmentation methods. Following standard protocols [41], we maintain the same dataset settings and data processing pipeline as the pretraining stage. For downstream training, we use 200 epochs with a learning rate of 0.001, a batch size of 24, and the Adam optimizer.

4.3.1. 3D Object Classification on Local Parts

We first train the classification model using our proposed momentum contrast pretraining method on five selected ShapeNet categories. The pretraining achieves a best top-1 accuracy of 51.42% at epoch 132, with the entire pretraining process requiring approximately 6.5 hours.

We then conduct two comparison experiments for 3D local parts classification. The first baseline trains a randomly initialized PointNet++ classification model to classify local clusters from the five ShapeNet categories. The second experiment fine-tunes the pretrained PointNet++ model obtained from the best momentum contrast pretraining run. The classification results are presented in Table 3.

After 200 epochs of downstream training, both models achieve comparable final performance. However, the pretrained model reaches its best performance significantly faster, achieving optimal results at epoch 168 with a total training time of 3 hours and 38 minutes, compared to epoch 199 and 4 hours and 19 minutes for the baseline model. This represents approximately 16% reduction in training time to reach comparable accuracy.

The accelerated convergence with pretraining can be attributed to the initialization provided by momentum contrastive learning. During pretraining, the model learns to organize local features into a structured latent space where geometrically similar local regions are pulled together while dissimilar regions are pushed apart. This pretraining phase establishes a meaningful initial feature representation that captures fundamental local geometric patterns. When fine-tuned for downstream classification, the classification head can leverage this structured feature space, enabling faster convergence compared to learning from random initialization. These results demonstrate that our momentum contrastive pretraining method successfully learns transferable local features from 3D point clouds, accelerating downstream training while maintaining competitive final accuracy. The comparable final performance indicates that the pretrained features do not sacrifice representational capacity, while the reduced training time validates the practical utility of local feature pretraining for 3D understanding tasks.

Table 3. Comparison Result of 3D Object Classification on Local Parts

	w/ Pretrained Model	w/o Pretrained Model
Best Instance Accuracy [%]	81.35	81.57
Best Class Accuracy [%]	75.70	76.08
Training Time Cost	3h 38m	4h 19m
Epoch Number for Best Result	EP168	EP199

5. Discussion

Our experimental results demonstrate that momentum contrastive learning can be successfully adapted to learn discriminative features from local point cloud regions, addressing a fundamental limitation of existing self-supervised methods that focus exclusively on complete object shapes. The finding that local clusters must contain at least 30% of the complete object to enable effective contrastive learning provides important insight into the information requirements for local feature learning. This threshold suggests that while local geometric patterns are crucial for object recognition, a certain minimum amount of structural context is necessary to form stable positive pair relationships in contrastive learning. This finding aligns with observations from previous studies on 3D object recognition under occlusion, which have shown that performance degrades significantly when less than 30% of an object is visible. Our work provides a principled approach to learning from such partial observations through self-supervised pretraining, offering a potential solution to the challenge of recognizing objects in cluttered real-world environments.

The effectiveness of random input dropout and random translation as augmentation methods, combined with the minimal contribution of random rotation, reveals important characteristics of local geometric feature learning. The critical role of dropout suggests that learning to recognize partial patterns is essential for robust local feature representations, as this augmentation directly simulates the occlusions and missing data encountered in real-world 3D sensing. The importance of translation augmentation indicates that position-invariant features are valuable for local pattern recognition, encouraging the model to focus on intrinsic geometric relationships rather than absolute spatial positions. In contrast, the negligible impact of rotation suggests that small local patches exhibit inherently weak directional characteristics, making rotation augmentation redundant for local feature learning. These findings differ from previous work on complete object recognition, where rotation augmentation typically provides substantial benefits, highlighting that local feature learning requires a distinct augmentation strategy tailored to the properties of local geometric patterns.

The downstream classification results, showing comparable final accuracy but significantly reduced training time with pretraining, provide evidence that momentum contrastive learning successfully captures transferable local geometric knowledge. The accelerated convergence can be attributed to the structured latent space established during pretraining, where geometrically similar local regions are organized into coherent clusters. This pretraining phase provides an informative initialization that enables the downstream classification head to learn more efficiently, as it can leverage existing local feature representations rather than learning from random initialization. However, the comparable final accuracy between pretrained and non-pretrained models suggests that the primary benefit of our approach lies in training efficiency rather than ultimate representational capacity. This finding indicates that while local feature pretraining accelerates learning, the downstream task eventually acquires comparable local feature representations through sufficient training iterations, suggesting that pretraining is most valuable in scenarios with limited computational resources or when rapid model deployment is required.

Future research directions include extending our framework to more complex downstream tasks beyond classification, such as 3D object detection and semantic segmentation, where local feature understanding plays an even more critical role. Additionally, investigating the scalability of our approach to larger and more diverse point cloud datasets, particularly real-world scanned data with significant noise and occlusion, would provide valuable insights into the practical applicability of local feature pretraining. The integration of our local feature learning framework with recent advances in transformer-based architectures and multi-modal learning represents another promising direction, potentially enabling more powerful and generalizable 3D representations. Finally, exploring the combination of local and global feature learning within a unified self-supervised framework could yield synergistic benefits, allowing models to capture both fine-grained geometric details and broader structural context simultaneously.

6. Conclusion

This work proposed a momentum contrastive learning framework specifically designed for learning 3D local feature representations from point clouds, addressing the previously unexplored problem of self-supervised learning directly from partial point cloud observations. Building upon the MoCo architecture, we adapted the momentum-based contrastive mechanism to process randomly sampled local clusters and designed a data processing pipeline combining random input dropout, random translation, and local KD-Tree sampling to generate diverse local clusters that simulate real-world partial observation scenarios. Systematic experiments on ShapeNet demonstrated that local clusters containing at least 30% of complete objects are necessary for effective contrastive training, with random input dropout and random translation proving essential for learning robust local features. Downstream classification experiments validated that models pretrained with our approach achieve comparable final accuracy to models trained from scratch while reducing training time by approximately 16%, confirming the effectiveness and transferability of our learned local representations. This work demonstrates that momentum contrastive learning can be successfully adapted to learn discriminative features from local point cloud regions, providing a simple and effective approach for local geometric feature learning that benefits downstream 3D understanding tasks through improved training efficiency.

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